

Epic: Deployment

Project: LegalEase

1. Epic Overview

Deploy LegalEase to production through a secure, automated and repeatable process, with the infrastructure, monitoring and rollback plans needed to run it reliably.

2. Background & Problem Statement

Once the model, architecture and APIs are ready, the product must reach users safely. Manual or inconsistent deployments risk downtime, configuration errors and exposure of confidential legal data.

3. Goals & Objectives

- Set up dev, staging and production environments
- Automate build, test and release with a CI/CD pipeline
- Meet security, data residency and compliance requirements
- Provide monitoring, alerting and quick rollback

4. Stakeholders

Role	Interest
Product Owner	Release timing, go-live approval
DevOps / Platform Engineers	Infrastructure, pipeline, releases
Backend / ML Engineers	Service and model deployment
Security / Compliance	Access control, data residency, audit
QA	Release validation

5. Scope

5.1 In Scope

- Environment setup (dev, staging, production)
- Infrastructure as code
- CI/CD pipeline and release process
- Secrets and configuration management
- Monitoring, logging, alerting and backups
- Rollback and disaster recovery plan

5.2 Out of Scope

- New feature development
- Model selection and training
- Long-term cost optimization

6. Assumptions & Constraints

- Hosting approach (cloud or self-hosted) is decided during the architecture epic
- Client data must stay within approved regions
- Production access is limited and audited

7. Deployment Environments

Environment	Purpose
Development	Day-to-day building and testing by engineers
Staging	Production-like validation before release
Production	Live system used by customers

8. Deployment Flow

- 1 Code merged and build triggered
- 2 Automated tests and security scans run
- 3 Build packaged and deployed to staging
- 4 Validation and approval
- 5 Release to production (gradual rollout)
- 6 Post-release monitoring; rollback if needed

9. Non-Functional Requirements

- Availability: target uptime defined and monitored
- Security: encryption, least-privilege access, secret management
- Scalability: handle expected load and traffic spikes
- Recoverability: backups and tested recovery procedures

10. User Stories

- 1 As a DevOps engineer, I want infrastructure defined as code so that environments are consistent and repeatable.
- 2 As a developer, I want an automated pipeline so that every change is built, tested and deployed the same way.
- 3 As a product owner, I want staged releases with easy rollback so that a bad release does not affect all users.
- 4 As a security lead, I want secrets and client data protected in every environment.
- 5 As an operations lead, I want dashboards and alerts so that issues are detected before customers notice.

11. Acceptance Criteria

- Dev, staging and production environments created and documented
- CI/CD pipeline deploys to staging and production successfully
- Rollback tested and completed within the agreed time
- Monitoring and alerts active for key services
- Security and compliance checklist signed off

12. Deliverables & Milestones

Phase	Deliverable
1. Plan	Deployment strategy and environment design
2. Setup	Infrastructure and CI/CD pipeline
3. Validate	Staging deployment, load and security testing

Phase	Deliverable
4. Go-live	Production release with rollback ready
5. Stabilize	Monitoring review and runbook handoff

13. Risks & Mitigation

Risk	Mitigation
Failed or faulty release	Staged rollout, automated rollback
Configuration drift between environments	Infrastructure as code
Secrets or data exposure	Secret manager, access control, audits
Unexpected load or cost spikes	Autoscaling limits, cost alerts

14. Dependencies

- Completed model, architecture and API epics
- Cloud or hosting account and budget approval
- Security and compliance review

15. Success Metrics

- Deployment success rate
- Deployment time
- Uptime
- Mean time to recover

16. Open Questions

- Cloud-hosted or self-hosted deployment?
- Which regions are required for data residency?
- Gradual rollout or full release at launch?